TEXORA · LECTURE SHEET

# Comfort: Definition, Types and Perception

*What clothing comfort is, its four recognised types — thermophysiological, sensorial, body-movement and psychological — how the skin's receptors perceive each, and the heat-balance physics that puts numbers on thermal comfort.*

AM-4103 · CLOTHING COMFORT & SPECIAL APPAREL PRODUCTS · TOPIC 1 · LEVEL 4 TERM 1

## 1. Defining comfort

**Comfort** is most usefully defined as *a pleasant state of physiological, psychological and physical harmony between a human being and the environment* (after Slater). Two features of the definition matter for engineering. First, comfort is largely the **absence of displeasure**: the wearer notices clothing only when something is wrong — too hot, too damp, scratchy, tight — so comfort engineering is mostly the systematic removal of negatives. Second, comfort is **perceived, not measured directly**: instruments measure fabric properties (Topic 2), but comfort itself happens in the wearer's nervous system, which is why this course pairs every instrument with the physiology of Topic 5.

The classic summary is that a comfortable garment keeps the body's microclimate — the thin layer of air and moisture between skin and fabric — near the comfort zone: skin temperature about **33–35 °C**, microclimate relative humidity below about **60%**, and no liquid sweat accumulating on the skin. Everything measurable in this course is a proxy for those three numbers.

## 2. The four types of comfort

| TYPE | WHAT IT COVERS | GOVERNING FABRIC/GARMENT PROPERTIES |
|---|---|---|
| Thermophysiological | Heat and moisture balance: the garment must pass metabolic heat and sweat vapour at the rate the body produces them, in the climate worn. | Thermal resistance (tog/clo), water-vapour resistance (Ret), air permeability, moisture absorption and wicking, thickness and trapped air. |
| Sensorial (tactile) | Everything the skin feels on contact: softness, smoothness, prickle, cling, wetness sensation, | Surface friction and roughness, fibre stiffness and |

| | | |
|---|---|---|
| | static. | diameter (prickle), hairiness, bending stiffness, surface moisture, electrostatic charge. |
| Body-movement (ergonomic) | Freedom to move: weight, fit, stretch and recovery, pressure on the body, garment dimensions matched to body motion. | Fabric stretch and elastic recovery, garment ease and pattern geometry, mass per unit area, seam placement, pressure (e.g. hosiery, compression wear). |
| Psychological (aesthetic) | How the wearer feels about the garment — fashion, colour, identity, occasion, brand, cultural norms. | Appearance, drape, colour and design, handle as judged visually, conformity to the wearer's self-image. |

The types interact. A fabric can pass every thermal test and still fail in the market because it feels prickly (sensorial) or looks cheap (psychological); a compression garment deliberately sacrifices "loose" comfort for controlled pressure that the wearer perceives as support. Product development therefore specifies comfort per type, not as a single number.

## 3. How comfort is perceived: the receptor systems

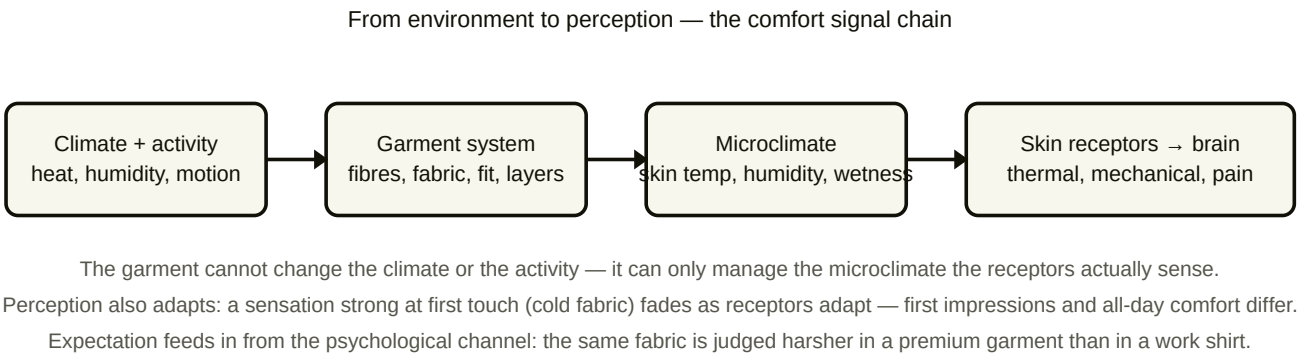



Figure 1 — The comfort signal chain. Fabric engineering works on link 2; wearer trials judge link 4.

The skin carries separate receptor families, and each maps to a comfort type. **Thermoreceptors** (warm and cold fibres) sense skin temperature and its rate of change — they drive thermal comfort and the "cold touch" shock of a fabric with high thermal absorptivity. **Mechanoreceptors** sense pressure, vibration and skin stretch — they read fabric contact, cling and movement restriction. **Nociceptors** (pain fibres) fire when stiff fibre ends (coarse wool, about > 30 µm diameter) bend against the skin — the physical basis of **prickle**. There is **no dedicated wetness receptor**: wetness is inferred by the brain from combined cold + touch signals, which is why a cold, smooth synthetic can feel "damp" when bone

dry. Comfort perception is thus an integration, weighted by adaptation, expectation and individual variation — the reason instrumental results (Topic 2) are always finally checked by wearer trials (Topic 4).

## 4. The physics: heat balance and the tog/clo system

Thermal comfort requires the body's heat balance (Topic 5) to hold without strain: heat produced = heat lost. Clothing resists the dry part of that loss, and the resistance is the engineer's design variable. Thermal resistance $R_{ct}$ is measured in m²·K/W and in two trade units:

$$1 \text{ tog} = 0.1 \text{ m}^2\cdot\text{K/W} \quad \cdot \quad 1 \text{ clo} = 0.155 \text{ m}^2\cdot\text{K/W} = 1.55 \text{ tog}$$

The **clo** is defined so that a resting person (heat production ≈ 58 W/m² of skin) is comfortable at 21 °C in 1 clo. Dry heat loss through the clothing follows:

$$q = (T_{skin} - T_{ambient}) \div R_{ct} \quad [\text{W/m}^2]$$

**WORKED EXAMPLES — VERIFY EVERY NUMBER**

**(a) Unit conversion.** A winter ensemble measures $R_{ct}$ = 0.186 m²·K/W. In tog: 0.186 ÷ 0.1 = **1.86 tog**. In clo: 0.186 ÷ 0.155 = **1.20 clo**.

**(b) Single fabric.** A fleece of thickness 2.0 mm and thermal conductivity k = 0.045 W/m·K has R = d ÷ k = 0.002 ÷ 0.045 = **0.044 m²·K/W = 0.44 tog** — fabric alone; the trapped air layers of a worn garment add most of the real insulation.

**(c) Heat loss.** Skin at 33 °C, air at 20 °C, clothing $R_{ct}$ = 0.155 m²·K/W (1 clo): q = (33 − 20) ÷ 0.155 = **83.9 W/m²**, or × 1.8 m² body area ≈ **151 W** total dry loss — more than a resting body produces (~107 W for 1.85 m² at 1 met), so at rest in 1 clo at 20 °C the wearer slowly cools: the arithmetic behind "put on another layer".

## 5. Moisture: the second half of thermophysiological comfort

Heat is only half the balance. Sweat that cannot evaporate wets the microclimate, collapses trapped air (insulation falls), makes fabric cling (sensorial comfort falls) and chills the wearer when activity stops. Fabrics handle moisture by three routes: **absorption** into hygroscopic fibres (cotton, wool — buffering vapour but slow to dry), **wicking** along fibre surfaces and capillaries (profiled synthetics — spreading liquid to evaporate), and **vapour diffusion** through pores and membranes (measured as Ret / MVTR, Topic 2). Comfort design chooses the route per product: next-to-skin sportswear wicks; a rain shell must diffuse vapour while blocking liquid rain — the central problem of protective clothing in Topics 3–4.

## Glossary

| TERM | MEANING |
|---|---|
| Microclimate | The air layer between skin and clothing, characterised by its temperature and humidity; the immediate environment the receptors sense. |
| Tog / clo | Units of thermal resistance: 1 tog = 0.1 m²·K/W; 1 clo = 0.155 m²·K/W = 1.55 tog. |
| Prickle | Pain-fibre response to stiff fibre ends (typically > ~30 µm) pressing on the skin; a sensorial-comfort fault of coarse wool. |
| Thermal absorptivity | Surface property governing the warm/cool feel at first contact; high values feel cool. |
| Wicking | Capillary transport of liquid moisture along fibres/yarns away from the skin. |

## Key points and summary

- Comfort is perceived, mostly as the absence of displeasure, and is judged in the microclimate: skin 33–35 °C, RH below ~60%, no liquid sweat film.
- Four types — thermophysiological, sensorial, body-movement, psychological — are specified and tested separately, and they interact.
- Perception comes from thermoreceptors, mechanoreceptors and nociceptors; wetness has no receptor and is inferred from cold + touch.
- Thermal resistance converts as 1 tog = 0.1 and 1 clo = 0.155 m²·K/W; dry heat loss is q = ΔT ÷ R.

## Practice

Answer in your own words; brief model answers follow.

1. A jacket system measures $R_{ct}$ = 0.248 m²·K/W. Express it in tog and clo.
2. At skin 34 °C and ambient 24 °C, what $R_{ct}$ limits dry heat loss to 50 W/m²? Express the answer in clo.
3. Classify by comfort type: (a) a shirt that feels scratchy; (b) a jacket that keeps the wearer warm but clammy; (c) trousers too tight to climb stairs; (d) a uniform the staff dislike because of its colour.
4. Why can a cold, dry polyester shirt feel damp at first touch?
5. Why does wet fabric insulate worse than the same fabric dry?

### Answer key

1. Tog: 0.248 ÷ 0.1 = 2.48 tog. Clo: 0.248 ÷ 0.155 = 1.60 clo.
2. R = ΔT ÷ q = (34 − 24) ÷ 50 = 0.20 m²·K/W = 0.20 ÷ 0.155 = 1.29 clo.
3. (a) Sensorial. (b) Thermophysiological (moisture side). (c) Body-movement/ergonomic. (d) Psychological.
4. There is no wetness receptor; the brain infers wetness from cold + smooth-touch signals. Cold polyester with high thermal absorptivity delivers exactly that signal pair, dry or not.
5. Water displaces trapped air (the real insulator) and conducts heat about 20+ times better than air; wet fibres also collapse loft, thinning the layer. Insulation falls on both counts.

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